Gaurav Vanvari

Software Engineer

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Experienced and results-oriented Software Engineer with a track record of 7 years in development and testing. Dedicated to ensuring the reliability and quality of applications to deliver exceptional customer experience.

Professional Experience

Bloomberg LP | Princeton, NJ — Sep 2020 - Present

Bloomberg's EMS manages end-to-end workflows, enabling dynamic control of positions and portfolios

Software Engineer

- Development: Developed scenario management and status reporting tool using Typescript, R+, and Comdb2 to enhance test coverage and categorize test failures
- Design and Architect: Designed and developed backend testing application and its APIs using Python, Docker, Jenkins, Alexandria, and Sphinx. The published API to the artifactory increased cross-team collaboration to eliminate effort duplication and made EMSX more reliable
- UI Test Automation: Automated 110 UI workflows with TypeScript, BDD and Cucumber, cutting regression testing time by 30%. Automated daily builds with Jenkins on Windows VMs in Azure and Linux machines in Bloomberg Cloud, ensuring continuous delivery. Managed Bloomberg Cloud environments for storing and testing build artifacts, enhancing debugging and compliance
- Test Suite Maintenance: Developed Security Price Tracker and Proactive Alerting System using Python to automatically update feature files with changing security prices and to report expiring securities, minimizing disruptions to test suite execution
- Hackathons: Created a Python machine learning project to automate ticket creation from error message logs
- Project Management and Mentorship: Served as the Scrum Master for the team and fostered knowledge sharing, product awareness, and team growth through trainings and demos
- Compliance: Contributed to EMSX's effort to be Digital Operational Resilience Act (DORA) compliant

Environment: *Behavioral Driven Development (BDD), Python, TypeScript, Cucumber, Gherkin, Jenkins, Bloomberg Terminal, GitHub, Grafana, GUTS, Splunk, Humio, NodeJS, Docker, Azure, Chromium*

IronNet Cybersecurity | Tysons Corner, VA — Sep 2017 - July 2020

IronDefense, IronNet's Next-Generation AI/ML Network Traffic Analysis (NTA) solution, detects network anomalies using machine learning and threat intelligence along the Cyber Kill Chain.

Software Engineer in Test

- Development: Developed and tested IronDefense API's and integrations with various tools (CrowdStrike, Phantom, QRadar, ServiceNow, Splunk, Swimlane, and XSOAR) to expedite security alert triage using Go, Python, Typescript, and Docker for incident response
- Web Application Security: Configured and deployed Snyk to GitHub repos, helping the engineering team identify and fix security vulnerabilities in the OSS supply chain
- Compliance: Collaborated with cross-functional teams to get IronDefense FedRAMP certified
- Tested queries in collaboration with Threat Hunters to test Hunt Platform and Splunk
- UI Test Automation: Automated 15% of manual tests through Cypress.io, boosting regression test runs tenfold and optimizing efficiency

- Backend Test Automation: Automated functional testing by replacing TestRail test cases with Go, gRPC Postman for nightly TeamCity runs
- ML Model Validation: Generated 13 network traffic anomalies via Ixia for IronDefense validation
- Chaos Testing: Conducted proactive Chaos Testing by leveraging Grafana to simulate service failures, successfully averting 12 high-impact chaos failures prior to release
- Load and Soak Testing: Benchmarked and certified 1G, 5G, and 10G IronDefense sensors to ensure consistent data ingestion and processing
- Manual Testing: Reported and validated over 200 bugs and 700 tasks/bugs throughout 20 IronDefense releases, contributing to product stability and quality
- Collaboration: Authored sales, customer, and SRE documentation, ensuring a seamless deployment process, and supported the Sales team in delivering effective IronDefense demonstrations to partners

Environment: *Agile, AWS, CitusDB, Cypress, Docker, Git, Go, gRPC, Grafana, Ixia, Java, Jira, Javascript, Kubernetes, Postgres, Postman, Python, REST API, TeamCity, TestRail, cURL, EDR, ITSM, SIEM, and SOAR*

Cogent Infotech | Salem, OR May 2017 – Sep 2017

Software Engineer

- Developed batch jobs such as Assistance Status Change, Assess Annual Service Free, Process Suspended Collections, and Electronic Payment Withdrawal using Java and Hibernate
- Developed Java-based Batch Jobs to trigger and display results for workers and case managers

Environment: *Hibernate, HQL, IBM (DB2, Jazz, RTC, Data Studio), Java, Jira, SonarLint, XML*

MediaTek USA Inc. | San Jose, CA May 2015 - Aug 2015

Software Engineering Intern

- Created a packet Generator that generates Architected Queuing Language (AQL) packets in accordance with Heterogeneous System Architecture and OpenCL 2.0 specifications using C++
- Implemented a Queue for efficient buffering of CPU-generated packets to optimize GPU processing

Environment: *C++, HSA, OpenCL, Perforce*

Certification and Training

API Security Foundations, APIsec University April 2024

OWASP API Security Top 10

Security+, CompTIA Dec 2023

IAM, Wireless Networks and Attacks, Malwares, Application Vulnerabilities and Attacks, Risk Management, Protecting Backups, Cryptography, Hashing, Data Classification, PKI

Jr Pentester , TryHackMe Oct 2023

Owasp top 10, Burp Suite, nmap, Metasploit, John the Ripper, Gobuster, Hydra, enum4linux, SQLite, Msfvenom

Certified Cloud Practitioner, AWS Jan 2020

Availability Zones, AWS CloudTrail , Amazon EC2, AWS Billing Console, Amazon S3, Route53 , IAM

Academic Projects

FPGA vs GPU performance comparison for Digit Recognition, NYU Spring 2016

- Used GPU and FPGA to speed up Convolution and Max Pooling layers, achieving a 13x speedup for Convolution and 5x for Max Pooling on GPU, when training and testing with the SVHN dataset

Advanced Encryption Standard, NYU Fall 2015

- Enhanced AES-256 design by utilizing Spartan-6 on-chip Digital Clock Manager and Block RAM to generate clock and store keys, leading to improved speed and reduced space
- Enhanced security by adding an authentication module, allowing only approved users to modify the key

RC5 Encyrption Algorithm, NYU Fall 2014

- Implemented RC5 block cipher with 64-bit input, 12 rounds, a 16-byte user key, and an accompanying RC5 Round Key Generation module with 26 words and 12 rounds using VHDL and Spartan-6 FPGA
- Validated correctness with a VHDL Testbench for 10,000 test vectors, cross-verifying results using Python

Education

New York University | Brooklyn, NY May 2016

Master of Science in Computer Engineering, GPA: 3.4/4.0

Indraprastha University (GGSIPU) | New Delhi June 2014

Bachelor of Technology in Electronics and Communications Engineering, 3.5/4.0